1. Summarize in 5 points Big Data?What are the Five V’s of Big Data
2. Identify the technological challenges in Managing Big Data
3. Explain Big data Ecosystem
4. Outline the characteristics of Big Data
5. Identify the three major sources of Big Data?Describe
6. Explain with suitable diagram the Big Data processing architecture
7. Illustrate how Big Data Problems are handled by Hadoop?
8. Summarize what is structured, semi structured and unstructured data?give example and explain
9. Explain its physical architecture of Hadoop wih diagram
10. State limitations of Hadoop.
11. Explain Hadoop ecosystem with core components,
12. Compare big data Analytics with traditional data mining and warehousing systems
13. Give any two examples of big Data case studies and illustrate which V's are satisfied by these case studies.
14. Explain Hadoop.Why do we need Hadoop
15. Classify and explain the core components of Hadoop with diagram
16. Give a brief overview of hadoop
17. Compare between traditional RDBMS and Hadoop
18. Explain disributed File System in context of Big data scenario
19. Outline HDFS and what are its features?
20. Explain the following terms NameNode Data Node.
21. Why is HDFS more suited for applications having large data sets and not when there are small files?
22. Explain the following terms Job tracker, Task Tracker, block in HDFS?
23. Explain Map Reduce? Explain how do map and reduce work
24. Identify what sorts of actions does the Job Tracker process perform?
25. Identify how does NameNode and dataNode tackle failures?
26. Explain Input Split in MapReduce and shuffling in Map Reduce?
27. Explain what are combiners? When should one use a combiner in MapReduce Job?
28. Model how does a master node differ from the worker node?
29. Identify in the context of key value pair data format, what are its benefits? what are its limitations?
30. "Write pseudo code for matrix vector multiplication by MapReduce.Illustrate with an
31. example showing all the steps"
32. Explain MapReduce? Explain role of combiners with the help of an example.
33. List the advantages and limitation of Hadoop
34. Summarize the Hadoop ecosystem components with diagram
35. List the limitations of Hadoop
36. Explain Union, Intersection and difference relational operations with pseudo code of Map and Reduce
37. List Relational Algebra operations and explain any two using MapReduce.
38. When it comes to big data, how does NoSQL score over RDBMS?
39. Identify the variations of NoSQL architectural patterns?
40. Give an example of shared and shared nothing.
41. Explain four ways by which big data problems are handled by NoSQL.

42. Explain NoSQL database? What are the different kinds of NoSQL databases?what are the advantages of each?
43. Compare and contrast between HBase, Cassandra and MongoDB?
44. "identify the different architecture patterns in NoSQL? Explain Graph Data store and
45. column family store patterns with relevant examples."
46. Explain NoSQL? Explain CAP Theorem
47. Agility is the NoSQL Business driver. Justify.
48. Explain in detail a short note on Casssendra.
49. Compare with 5 points of difference between SQL and NoSQL
50. Explain in detail DynamoDB
51. Explain the Image compression NoSQLsolution for big data
52. List the application of autoencoders
53. What are stream queries? Explain different categories of stream queries
54. What are the different issues in data stream processing
55. List and explain various data stream sources
56. What is sampling of data in a stream? How do we obtain representative sampling
57. What are the different types of sampling of data in a stream?
58. Comment on space requirements in count distinct problem
59. How frequent items in a stream are counted
60. Explain the filtering process of data stream with suitable examples
61. Compare between standing queries and AdhocQueries
62. Describe any two sampling techniques for big data with the help of examples
63. Explain abstract architecture of DSMS
64. Explain the concept of Bloom'sFilter using an example
65. Explain count distinct problem with suitable example
66. Explain the process of combining the Estimates. Also comment on space requirements
67. Using an example bit stream explain the working of the DGIM algorithm to count the number of 1's (Ones) in a stream
68. What are Decaying Windows? Explain in detail?
69. Explain the DGIM algorithm
70. Explain Flajolet Martin algorithm in detail
71. What are distance measures? Briefly any two distance measures
72. What do you mean by Euclidean Distance? Explain with example
73. What is Cosine distance? Explain with suitable example
74. What is Edit distance? Explain with classical method
75. Consider the following two vectors in Euclidean space X=[1,2,-1] and Y=[2,1,1]. Calculate the Cosine distance between X And Y
76. Write pseudo function for cluster algorithm
77. Explain different criteria regarding distance measures
78. Explain with Longest Common Sequence(LCS) method
79. Explain Hamming distance with an example
80. Explain CURE algorithm for large scale clustering
81. Write procedure for merging cluster in CURE
82. What is bucket, how it is used for clustering?
83. Explain in brief initialising and merging of bucket
84. Define the graph parameters listed Below I. Density II. Centrality
85. Define the graph parameters listed below: I. Degree II.Geodesic distance

86. Give the mathematical formulation of degree of centrality, Closeness of Centrality Between's centrality
87. Differentiate between Collaborative and Content Based Filtering
88. State algorithms for finding communities in a social graph
89. What are back links and forward links with suitable examples
90. What is thrashing? How does it affect the Page ranking mechanism?
91. Explain Page Rank with an example. Explain importance of Page Rank.
92. List down the steps in modified Page Rank Algorithm
93. Explain Iterating Pagerank Process with Map Reduce. Also comment on the use of Combiners
94. Explain utility matrix with example
95. Explain with example Content Based filtering in a recommendation system
96. Explain with example Collaborative filtering in a recommendation system
97. Explain the different modification suggested inefficient computation of Web Pages
98. Explain item profile of content based recommendation
99. Explain in details CollaborativeNetwork, Email Network andTelephone Network in detail
100. Explain Hierarchical clustering algorithm
101. Explain Fuzzy C-means clustering algorithm
102. Explain the process of distance measurement in the Social-Network Graph
103. Write a note on Social Network Graph
104. Differentiate between different clustering algorithms